



Adaptation of thermal reaction norms in constant and fluctuating environments

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Received: 30 August 2024 / Accepted: 9 March 2025 / Published online: 25 March 2025
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Abstract

Thermal reaction norms depict how temperature influences biological performances, thus also known as thermal performance curves (TPCs). Arguably, the interplay of the thermal environment and the TPC can shape the strength of natural selection, thereby driving the long-term evolution of the TPC. We develop a Lotka-Volterra model, using adaptive dynamics (AD), to investigate how constant versus periodically fluctuating environmental temperatures drive the TPC adaptation. To formulate invasion fitness under fluctuating selection, we assume that the intrinsic rate of growth and carrying capacity to be temperature dependent, and also that the competition coefficient from one individual to another is proportional to the ratio of their beta-distribution-shaped thermal performances at the current environmental temperature. Results show that, under a constant temperature, the optimal temperature of the TPC evolves to align perfectly with the environmental temperature, with the TPC breadth shrinking to zero, reflecting local adaptation to complete thermal specialisation. In fluctuating thermal environments, adaptation produces broader TPCs, with their optimal temperature potentially mismatching the average environmental temperature. When the TPC's optimal temperature matches the average temperature, large temperature fluctuations lead to broad TPCs (thermal generalisation). Our model also shows the emergence of bimodal TPCs under rapid and large temperature fluctuations, indicating adaptation to extreme temperatures and potentially a divergence of thermal strategies within the population. Our theoretical model has demonstrated that adaptation of TPCs in periodic thermal regimes promotes the evolution of thermal generalists and possible character divergence, compared to complete thermal specialisation in constant environments.

Keywords Adaptive dynamics · Beta distribution · Optimal temperature · Performance breadth · Thermal generalisation · Thermal specialisation

Introduction

Temperature is a fundamental abiotic factor that directly influences the performance of ectotherms, impacting their physiological rates such as fecundity, growth, and metabolic rate that are crucial parts of the life cycle (Angilletta 2009; Mallard et al. 2020; Schulte et al. 2011). Thermal reaction norms, also known as thermal performance curves (TPCs), describe how temperature affects these biological processes (Schulte et al. 2011). TPCs are commonly utilised to capture the responses of organisms or populations to thermal changes (Sinclair et al. 2016). They typically exhibit a unimodal shape, demonstrating similar asymmetric and skewed responses to different temperatures (Angilletta 2009; Schulte et al. 2011; Sinclair et al. 2016). These curves often depict a gradual increase in performance as the temperature rises, reaching the optimal temperature where maximal performance is achieved, followed by a sharp decline (Angilletta 2009; Huey and Kingsolver 1989; Izem and Kingsolver 2005). TPCs are bounded by the critical minimum and maximum temperatures, which define the thermal tolerance range. A smaller tolerance range generally represents a narrower TPC breadth, while a larger tolerance range a wider TPC breadth (Sinclair et al. 2016; Stevenson et al. 1985).

Evolutionary adaptation to different environmental conditions can reshape the TPC by changing the height (through a vertical shift in performance), position (through a horizontal shift in optimal temperature), and the breadth (simultaneous vertical and horizontal shift) of the curve (Angilletta 2009; Izem and Kingsolver 2005; Knies et al. 2006; Schulte et al. 2011). Physiological constraints can impose a generalist-specialist trade-off between the TPC breadth and height, altering the temperature range over which organisms can perform efficiently, while keeping the area under the TPC constant (Angilletta 2009). Several theoretical studies have examined TPC evolution in fluctuating environments (Dowd et al. 2015; Estay et al. 2014; Kingsolver and Gomulkeiwicz 2003). However, no study has considered how interactions among individuals with different TPC forms—varying in optimal temperature and breadth—might affect competitive outcomes in fluctuating environments. These interactions, along with other temperature-dependent rates, could shape selective pressures and drive evolutionary adaptation to different thermal regimes.

Thermal fluctuations impose direct impacts on biological systems (Dowd et al. 2015; Jensen 1906; Ruel and Ayres 1999). Due to Jensen's (1906) inequality for a concave-down function, which is the typical shape of a TPC, thermal fluctuations drive the TPC's optimal temperature to exceed the average environmental temperature (Amarasekare and Johnson 2017; Pásztor et al. 2000). Using this hypothesis, Amarasekare and Johnson (2017) explored the mechanism driving latitudinal variation in the relationship between optimal temperatures in reaction norms of fecundity and average environmental temperatures in seasonal environments. They found that large seasonal fluctuations produced broad TPCs, characteristic of temperate species, with optimal temperatures exceeding the average body temperatures of ectotherms. In contrast, weak seasonal fluctuations resulted in narrower TPCs, typical of tropical species, with optimal temperatures closely matching the average body temperature (Amarasekare and Johnson 2017; Gilchrist 1995).

Many experimental evolution studies demonstrate that TPC characteristics differ between constant versus fluctuating environments (Duncan et al. 2011; Foray et al. 2014; Kassen and Bell 1998; Reboud and Bell 1997; Schaum et al. 2022). While constant environments favour thermal specialists with steep, narrow TPCs, fluctuating environments select for thermal

generalists with broader TPCs (Duncan et al. 2011; Janzen 1967; Gilchrist 1995; Levins 1968). More specifically, frequent thermal fluctuations promote thermal generalists, while constant temperatures or slow thermal fluctuations favour specialists (Buckling et al. 2007; Kassen 2002; Kassen and Bell 1998). Previous theories and comparative studies suggest that thermal specialists, such as tropical ectotherms, are adapted to environments with minimal seasonal fluctuations. In contrast, thermal generalists, such as temperate ectotherms, are adapted to environments with large seasonal fluctuations (Gilchrist 1995; Huey and Kingsolver 1989; Levins 1968). These generalists tend to have optimal body temperatures that exceed the average environmental temperature (Amarasekare and Johnson 2017; Angilletta 2009).

Here, we investigate how environmental fluctuations can shape population dynamics and natural selection on the TPC through its effect on intraspecific competition (Amarasekare and Coutinho 2014; Amarasekare and Johnson 2017; Johnson et al. 2016; Laws and Belovsky 2010; Van Doorslaer et al. 2009). To this end, the adaptive dynamics (AD) framework is appropriate for modelling evolution of life-history traits, where ecological and evolutionary processes act synergistically to influence the outcome of trait-mediated and density-dependent ecological interactions (Dieckmann and Law 1996; Doebeli and Dieckmann 2000; Geritz et al. 1998). Specifically, we develop a Lotka-Volterra model that incorporates temperature-mediated population growth, carrying capacity, and intraspecific competition among organisms with different TPC shapes. Using adaptive dynamics, we study the resulting evolution of the TPC (optimal temperature and breadth) in different thermal environments. We investigate the following: (i) Can fluctuating environments cause the optimal temperature of the TPC to align with or deviate from the average environmental temperature, in contrast with the perfect alignment in the constant temperature environment? (ii) Do these temperature fluctuations lead to the evolution of thermal generalists (large TPC breadths) compared to the constant temperature environment? (iii) How do changes in thermal fluctuations influence the ecological and evolutionary outcomes of TPC adaptation?

Materials and methods

Population dynamics under fluctuating temperatures

To model the periodic temperature variation experienced by ectotherms in their environments, we use a sinusoidal function, $\tau(t) = \tau_0 + m \sin(\omega t)$, where $\tau(t)$ is a variable (hereafter τ for short) and represents the environmental temperature at time t , τ_0 represents the average ambient temperature, m represents the magnitude, and ω is proportional to the frequency f of periodic temperature fluctuations ($f = \omega / 2\pi$), respectively. The period of fluctuations, denoted as T , is given by $T = 2\pi / \omega$. For convenience, we selected parameters τ_0 and m to satisfy $(\tau_0 - m) > 0$ and $(\tau_0 + m) < 1$, so to restrict thermal oscillations between the critical minimum temperature $C\tau_{\min}$ (set to 0) and the critical maximum temperature $C\tau_{\max}$ (set to 1), see Fig. S1. With $m = 0$, the ambient temperature remains constant over time such that $\tau = \tau_0$.

The dynamics of population size N in a periodically fluctuating environment can be depicted by the following logistic equation: $dN/dt = r(\tau)N(1 - N/k(\tau))$. Here, peri-

odic fluctuations in the ambient temperature leads to periodic fluctuations in the intrinsic population growth rate r and carrying capacity k . The metabolic rates of ectotherms characterising these parameters typically exhibit nonlinear temperature-dependence, as described by the metabolic theory of ecology (Brown et al. 2004; Savage et al. 2004). As such, the intrinsic growth rate of ectotherms experiencing the fluctuating thermal regime is described using a saturation function of exponential type (See Eq. S3.2, Appendix S3; Fig. S2): $r(\tau) = r_0 \exp(-c/\tau) + r_1$, where r_0 represents the scaling constant of the growth rate, c represents the steepness of the growth rate, and r_1 represents the growth rate at zero temperature. The carrying capacity of ectotherms experiencing the thermal environment, is modelled using a Gaussian function centred at the intermediate temperature (See Eq. S3.3, Appendix S3; Fig. S3): $k(\tau) = k_0 \exp\left(-(\tau - 0.5)^2 / \sigma_k\right)$, where k_0 represents the scaling constant of the carrying capacity and σ_k measures the range of variation of the carrying capacity. The maximum carrying capacity is reached at the intermediate environmental temperature (when $\tau = 0.5$), and lower carrying capacity occurs when the temperature becomes extreme (towards 0 or 1). Under certain conditions detailed in Appendix S3, the population size N converges asymptotically to a periodic state, although not necessarily following the same frequency as the ambient temperature fluctuation (Fig. 1a–b).

Resident-mutant population dynamics

To model adaptation in TPCs, we assume that rare mutants M arise within the resident population N , with the TPC shape of the mutants differing slightly from that of the residents. The intrinsic growth rate and the carrying capacity functions of the mutants are assumed to be the same as those of the resident population. The resident-mutant population dynamics are modelled using the following Lotka-Volterra competition model (Ackermann and Doebeli 2004; Doebeli 2011; Doebeli and Dieckmann 2000; Kisdi 1999):

$$\begin{aligned} \frac{dN}{dt} &= r(\tau) N \left(1 - \frac{N + \alpha_{NM}(\tau) M}{k(\tau)} \right) \\ \frac{dM}{dt} &= r(\tau) M \left(1 - \frac{\alpha_{MN}(\tau) N + M}{k(\tau)} \right) \end{aligned} \quad (1)$$

where $\alpha_{MN}(\tau)$ denotes the temperature-dependent competition strength of residents on mutants and $\alpha_{NM}(\tau)$ mutants on residents, under ambient temperature τ . Specifically, the competition strength between residents and mutants is assumed to be given by the ratio of their TPCs at temperature τ (Appendix S2; Schoener 1974), $\alpha_{MN}(\tau) = \beta_N(\tau; \mu_N, s_N) / \beta_M(\tau; \mu_M, s_M)$ and $\alpha_{NM}(\tau) = \beta_M(\tau; \mu_M, s_M) / \beta_N(\tau; \mu_N, s_N)$, where the TPC is modelled with the beta distribution function, $\beta(\tau; \mu, s)$, with the mean $\mu \in (0, 1)$ and scaling parameter $s \in (0, \infty)$ that is proportional to the TPC breadth (Appendix S1). The effects of the difference in the means and the scaling parameters between residents and mutants on the competition strength are illustrated in Fig. S4.

Evolutionary dynamics

To study the adaptation in TPC in constant and periodic environments, we utilise the adaptive dynamics (AD) framework (Dieckmann and Law 1996; Geritz et al. 1998;

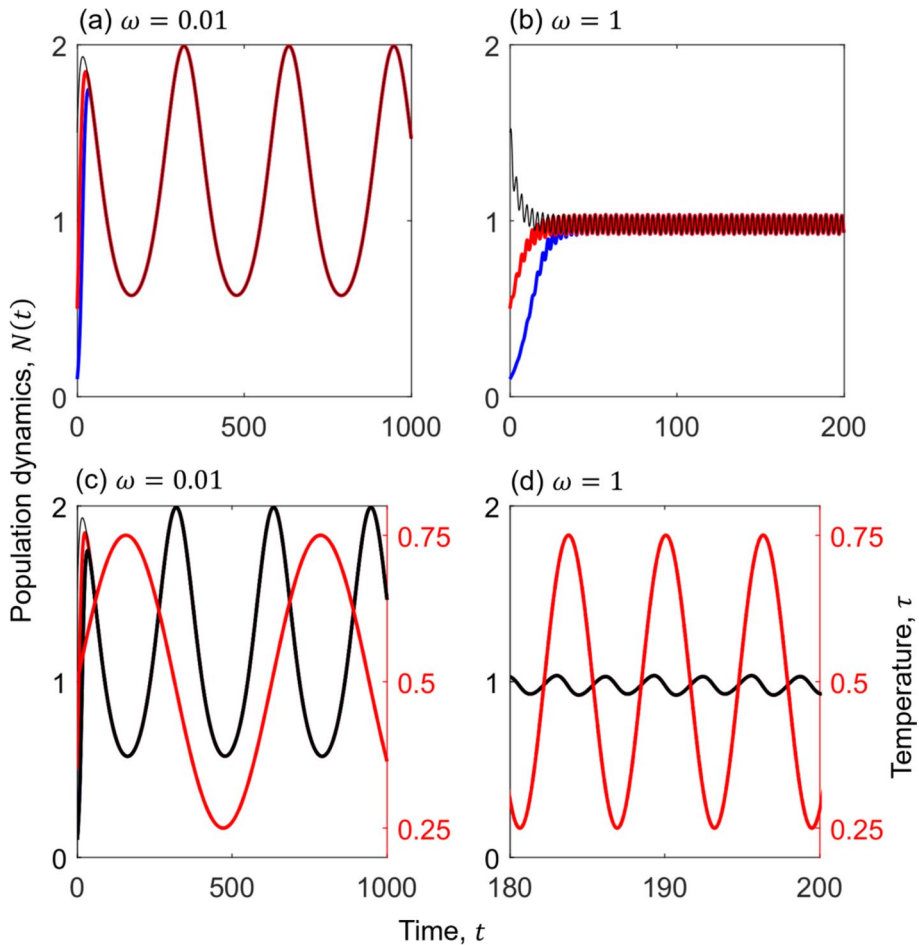


Fig. 1 Temperature and resident population dynamics. (a–b) The residents exhibit attracting periodic fluctuations with period 1 (1 maximum, 1 minimum, panel a) and period 2 (2 maxima, 2 minima, panel b), from different initial conditions. (c–d) Temperature dynamics are slower than population dynamics, with both slow ($\omega = 0.01$) and fast temperature fluctuations ($\omega = 1$). In both cases, the periodic temperature fluctuations and the population dynamics move in an asynchronous manner, producing a negative correlation between them. Parameters: $r_0 = c = r_1 = 0.1$, $k_0 = 2$, $\tau_0 = 0.5$, $m = 0.25$, and $\sigma_k = 0.05$. All model parameters are described in Table 1

Metz et al. 1992). Specifically, we depict the TPC shape using its mean (μ) and scaling parameter (s) proportional to its breadth. In fluctuating environments, the invasion fitness is determined as the long-term average per-capita growth rate of the rare mutant ($M \rightarrow 0$) within the resident population (Metz et al. 1992; Eq. S3.4, Appendix S3),

$$\bar{\lambda}(\mu_N, s_N, \mu_M, s_M) = \lim_{L \rightarrow \infty} \frac{1}{L} \int_0^L r(\tau) \left(1 - \frac{\alpha_{MN}(\tau) N(t)}{k(\tau)} \right) dt, \quad (2)$$

where L is the length of the time domain used for the long-term average; note, $\bar{\lambda}$ is a function of the TPC shape parameters of both the mutants and the residents (see pairwise invasibility plots in Fig. S5).

The selection gradients with respect to the mutant TPC mean and scaling parameter, denoted as $\bar{g}_\mu$ and $\bar{g}_s$, respectively, can be defined as the partial derivatives of the long-term invasion fitness with respect to the mutant TPC shape (μ_M, s_M), evaluated at the current resident TPC shape (μ_N, s_N) (See Eq. S3.10–S3.11, Appendix S3). Evolution proceeds through resident-mutant substitutions along the selection gradients. If $\bar{g}_\mu > 0$, mutants with a greater TPC mean will invade and substitute the residents. If $\bar{g}_\mu < 0$, the mutants with a greater TPC mean will be eliminated. Similarly, if $\bar{g}_s > 0$, mutants with a larger scaling parameter (wider TPC breadth) will invade and substitute the residents. If $\bar{g}_s < 0$, mutants with a larger scaling parameter will die out.

Assuming sufficiently small and rare mutations, the evolutionary dynamics of the TPC shape can be described by the canonical equations of AD (Dieckmann and Law 1996). These equations show that evolutionary rates are proportional to the product of the effective population size and the selection gradients (Eq. S3.13, Appendix S3). The evolution of the TPC reaches an evolutionarily singular point (μ^*, s^*) where the selection gradients vanish. At the singularity, directional selection disappears, and the second derivatives of the invasion fitness, given by the Hessian matrix, determine whether TPC evolution halts (reaching an evolutionarily stable strategy, ESS) or proceeds to initiate potential dimorphic evolutionary branching (Geritz et al. 2016; Della Rossa et al. 2015; Dercole et al. 2016) (Appendix S4).

Numerical analyses

To understand the ecological response of the population to ambient temperature fluctuations, we computed the bifurcation diagram of the population dynamics by simulating the logistic equation on a discrete grid of values for average ambient temperature τ_0 , magnitude m , and frequency ω of temperature fluctuations. In each bifurcation diagram, we tracked local extrema (maxima and minima) of the simulated population size (after discarding the initial transient phase of population dynamics) by varying one parameter while keeping others constant. We used the MATLAB ode15s function that is suitable for solving stiff differential equations.

To examine the effects of the periodic temperature fluctuation on the evolution of the TPC shape, we solved the eco-evolutionary model (Eq. 1, S3.13) using the MATLAB ode45 function based on the Runge-Kutta (4,5) formula with the Dormand-Prince pair (Dormand and Prince 1980). The time domain ranges from 0 to 2000 time units, to allow the population dynamics to asymptotically reach a periodic state before the invasion fitness is computed. To calculate the long-term average invasion fitness, we used the trapezoidal function in MATLAB, after excluding the transient phase (the first 100 time units of the population dynamics). The selection gradients were then computed on a discrete grid of μ and s values, from which the evolutionary nullclines corresponding to $\bar{g}_\mu = 0$ and $\bar{g}_s = 0$ were estimated, and the intersection of the two nullclines identified the evolutionarily singular point (μ^*, s^*).

From assessing the 2-dimensional evolutionary vector field in the (μ, s) plane, we confirmed the convergence stability of the singularity (also confirmed with the eigenvalues of the Jacobian matrix of the canonical equations). We further confirmed the evolutionary stability of the singularity by computing the eigenvalues of the Hessian matrices of the invasion fitness (for μ and s respectively). Finally, we evaluated the effect of the thermal regime, captured by average ambient temperature τ_0 , magnitude m and frequency ω of temperature fluctuation, on the evolutionary singularity, varying two parameters while holding the third parameter of the thermal regime constant.

Results

Ecological effects of temperature fluctuations

Periodic temperature fluctuations in the thermal regime can cause periodic fluctuations in population size, but not necessarily synchronised with nor at the same frequency as the thermal regime (compare the population fluctuations against slow ($\omega = 0.01$) or rapid ($\omega = 1$) temperature fluctuations in Fig. 1). The results indicate that, under either slow or fast temperature fluctuations, population size can fluctuate faster (i.e. a shorter cycle) than the temperature dynamics (Fig. 1c-d). In addition, the magnitude of population oscillations is greater for slow compared to rapid temperature fluctuations. The long-term local maxima and minima of the asymptotic population oscillations, which vary with the average ambient temperature (τ_0), are shown in the bifurcation diagrams (Fig. 2a-b). Under either slow ($\omega = 0.01$) or rapid ($\omega = 1$) thermal fluctuations, as the average ambient temperature τ_0 converges to 0.5 (centre), the period of population oscillations are halved (Figs. S6 and S7).

Under slow thermal fluctuation ($\omega = 0.01$), with the increase of the magnitude of temperature fluctuations (m) from zero (no fluctuation), the initial stable population size (equal to the carrying capacity $k_0 = 2$) forms the period-1 population oscillation (with a fixed and repeating cycle). As the magnitude of fluctuation increases, the maximum population size gradually decreases while the minimum population size declines to zero (Fig. S8a). Under fast thermal fluctuation ($\omega = 1$), the amplitude of the period-1 population oscillation becomes notably smaller, with both the maximum and the minimum population size declining rapidly to zero (Fig. S8b).

As the frequency of the thermal fluctuation increases from slow to rapid ($0.01 < \omega < 1$), the maximum population size of the period-1 population oscillation decreases despite an increase in the carrying capacity $k_0 = 2$. Meanwhile, the minimum population size increases, converging to a stable equilibrium through a Hopf bifurcation (Fig. S9). These findings highlight the non-linear response of population fluctuations and magnitudes to changes in thermal regimes, even without accounting for the adaptive evolution of TPCs. This provides valuable insights into the short-term impacts of seasonal fluctuations and climate change on population dynamics.

Evolutionary singularity under temperature fluctuations

By setting $m = 0$, the environmental temperature becomes constant ($\tau = \tau_0$). The TPC evolutionary singularity $(\mu^*, s^*) = (\tau_0, 0)$ is a continuously stable strategy that

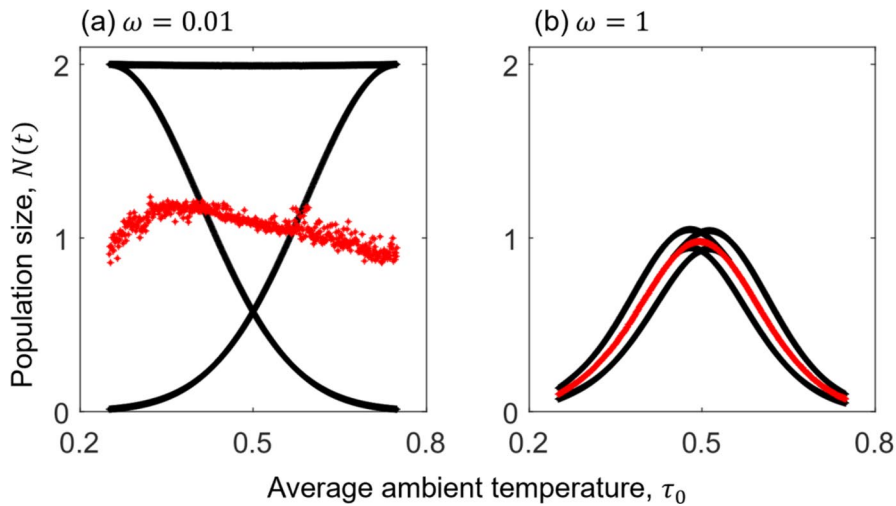


Fig. 2 Response of population dynamics ($N(t)$) to changes in the average ambient temperature (τ_0). The curves depict the relative maxima and minima of $N(t)$ to changes in τ_0 when it reached its periodic asymptotic state. (a) Under slow thermal fluctuations, as the average ambient temperature converges to $\tau_0 = 0.5$, the population oscillations transitions from period-2 (1 maximum and 2 minima) to period-1 (1 maximum and 1 minimum) cycles. (b) Under rapid thermal fluctuations, as the average ambient temperature τ_0 approaches the intermediate value, the population oscillations transition from period-2 (2 maxima and 2 minima) to period-1 (1 maximum and 1 minimum) cycles. Red curve: mean of the population cycles. Parameters: $r_0 = c = r_1 = 0.1$, $k_0 = 2$, $\tau_0 = 0.5$, $m = 0.25$, and $\sigma_k = 0.05$. All model parameters are described in Table 1

Table 1 Summary of the model parameters, their description and reference value(s)

Parameter	Description	Reference value
r_0	Scaling constant of the growth rate	0.1
c	Steepness of the growth rate	0.1
r_1	Growth rate at zero temperature	0.1
k_0	Scaling constant of the carrying capacity	2
σ_k	Width of Gaussian carrying capacity function	0.05
τ_0	Average ambient temperature	$\tau_0 \in (0,1)$
m	Magnitude of thermal fluctuations	$m \in (0,0.5)$
ω	Frequency of thermal fluctuations	$\omega \in (0.01,1000)$
μ	Mean of performance curve	$\mu \in (0,1)$
s	Scaling parameter of performance curve	$s \in (0,2)$

is both convergence stable and evolutionarily stable (Appendix S4). Under the fluctuating thermal regime ($m > 0$), for the specific parameters listed in Table 1, we observed essentially similar singularities, $(\mu^*, s^*) = (0.5, 0.14)$ for slow thermal fluctuation ($\omega = 0.01$) and $(\mu^*, s^*) = (0.51, 0.16)$ for fast thermal fluctuation ($\omega = 1$), both continuously stable (i.e., convergence stable [evident by the vector field indicating the selection gradients in Fig. 3] and evolutionarily stable).

Optimal temperature and breadth of thermal performance

To investigate how the TPC optimal temperature and breadth evolve in response to the thermal regime, we examined the effects of varying pairs of the environmental parameters characterising the thermal regime (average ambient temperature τ_0 , magnitude of oscillation m , and frequency of thermal fluctuation ω) on the optimal temperature (τ_{opt} , i.e., the temperature that maximises the TPC) and the TPC breadth (σ , i.e., the standard deviation of the beta function describing the TPC). These values are related to but distinct from the singular points (μ^* , s^*) of the TPC (see Appendix S1). The environmental variation leads to the emergence of both unimodal (region with colour gradient in Fig. 4) and bimodal TPCs (red region in Fig. 4).

We start by describing unimodal TPCs. First, under low-magnitude thermal fluctuation ($m \leq 0.1$), the optimal TPC temperature τ_{opt} aligns perfectly with the average environmental temperature τ_0 , but deviates when the magnitude m increases (Fig. 4a). The TPC breadth (standard deviation σ) enlarges with increasing thermal fluctuation magnitude m , and shrinks to zero with the decline of thermal fluctuation magnitude to $m = 0$ (Fig. 4b). Second, when the thermal environment fluctuates with moderate magnitude ($0.1 \leq m \leq 0.15$), the optimal TPC temperature deviates from the average environmental temperature (τ_0), evolving towards lower values if $\tau_0 < 0.5$ and higher values if $\tau_0 > 0.5$ (Fig. 4a). With the increase of the average environmental temperature (τ_0), the TPC breadth (σ) first increases, remains constant for the intermediate range when the optimal TPC temperature matches τ_0 , and then drops (Fig. 4b).

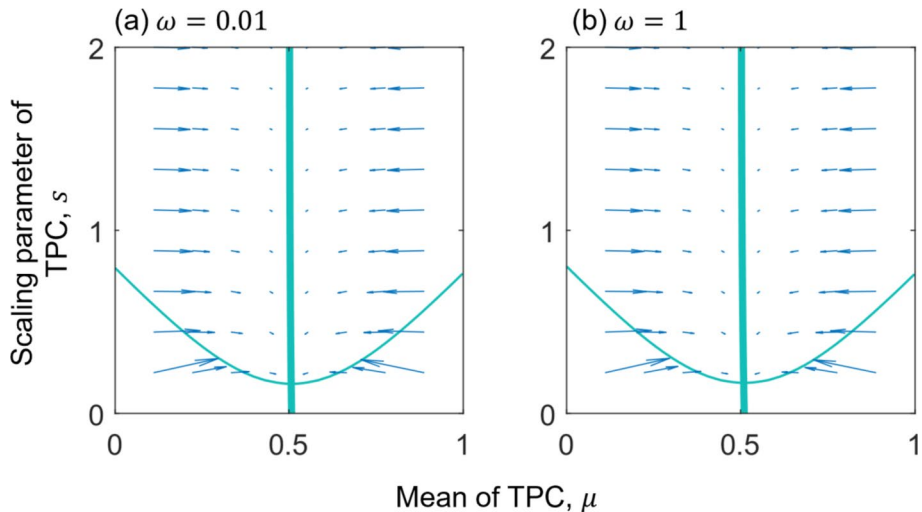


Fig. 3 Evolutionary vector fields of Eq. (9). Arrows indicate the magnitude and direction of evolutionary change. The bold curves represent the nullcline for the selection gradient of the mean of the TPC (i.e., $\bar{g}_\mu = 0$) and the thin curves represent the nullcline for the selection gradient of the scaling parameter of the TPC (i.e., $\bar{g}_s = 0$). The intersection of the nullclines determine the singular point, numerically solved as: (a) $(\mu^*, s^*) = (0.5, 0.14)$ for slow thermal regime ($\omega = 0.01$), and (b) $(\mu^*, s^*) = (0.51, 0.16)$ for fast thermal regime ($\omega = 1$). Both singular points are continuously stable (i.e., convergence stable, meaning the evolutionary dynamics converge towards it, and evolutionarily stable, meaning the end of the evolutionary process). Parameters: $r_0 = c = r_1 = 0.1$, $k_0 = 2$, $m = 0.25$, and $\sigma_k = 0.05$. All model parameters are described in Table 1

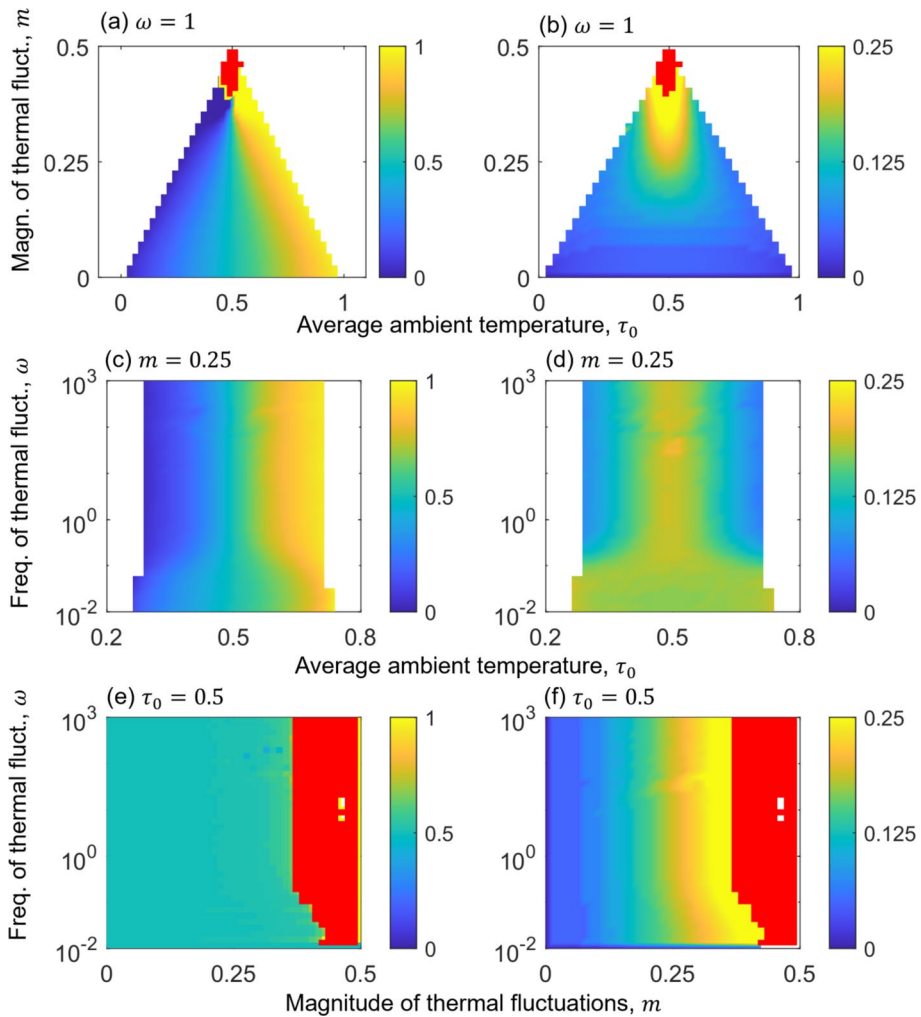


Fig. 4 Impact of two environmental parameters on the evolution of the optimal temperature (τ_{opt}) and standard deviation (σ) of the TPC, with the third parameter held constant. Investigating the effects of changes in: (a–b) average ambient temperature (τ_0) and magnitude of oscillation (m), while maintaining constant rapid fluctuations ($\omega = 1$), (c–d) average ambient temperature (τ_0) and frequency of thermal fluctuation (ω), while maintaining a constant intermediate magnitude of oscillation ($m = 0.25$), as well as (e–f) magnitude (m) and frequency of thermal fluctuation (ω), while maintaining an intermediate average ambient temperature ($\tau_0 = 0.5$). The coloured regions in each plot represent unimodal TPCs. The white regions in Figs. (a–d) represent unfeasible parameter zone (i.e., where τ_0 and m fail to satisfy $(\tau_0 - m) > 0$ and $(\tau_0 + m) < 1$), while the red regions represent bimodal TPCs. All model parameters are described in Table 1

Finally, under high-magnitude thermal fluctuation ($0.15 \leq m \leq 0.35$), the optimal TPC temperature converges to the intermediate average environmental temperature $\tau_0 = 0.5$ (Fig. 4a). Within this high magnitude thermal fluctuation region, the TPC breadth (σ) follows a bell shape along the average temperature axis, reaching the peak at $\tau_0 = 0.5$, although the monotonic increase of TPC breadth is clear with increasing magnitude of thermal fluctuation (Fig. 4b).

Bimodal TPCs also emerged but within a narrow parameter region, particularly in thermal environments with extremely large magnitude ($0.4 < m < 0.5$) and rapid temperature fluctuations (Fig. 4a–b). The bimodal TPC indicates that the population has adapted to the temperature extremes, implying a split of thermal strategies among individuals of the population (some adapted to one temperature extreme, and the rest to the opposite temperature extreme, leading to a bimodal TPC of the population). Although the beta distribution used to model the TPCs could express the extreme J shape, no such skewed TPCs were observed under the explored parameter ranges.

Next, we examine the effects of changing the average ambient temperature (τ_0) and the frequency of temperature fluctuation (ω) on the TPC shape, with the magnitude of thermal fluctuation set at an intermediate value ($m = 0.25$). These combinations of environmental parameters exclusively produced unimodal TPCs (Fig. 4c–d). Under slow temperature fluctuations ($0.01 < \omega < 0.1$), the optimal TPC temperature deviates from the average environmental temperature (Fig. 4c), with a consistently large TPC breadth (Fig. 4d). Within rapidly varying thermal environments ($1 < \omega < 1000$), the optimal TPC temperature evolves to a lower value than the average temperature when $\tau_0 < 0.5$ and a higher value when $\tau_0 > 0.5$ (Fig. 4c). With the increase of the average ambient temperature, the TPC breadth increases first and then drops, peaking at $\tau_0 = 0.5$. In contrast, moving from slow to fast thermal environments, the TPC breadth declines if the average ambient temperature is different from 0.5, but the TPC breadth increases for $\tau_0 = 0.5$ (Fig. 4d).

Finally, we assess the impact of changing the magnitude (m) and frequency of thermal fluctuation (ω) on the TPC shape, under the intermediate average ambient temperature ($\tau_0 = 0.5$). Both unimodal TPCs (region with colour gradient in Fig. 4e–f) and bimodal TPCs (red region) were observed. When considering the unimodal TPCs, varying both environmental parameters resulted in the optimal TPC temperature maintaining a constant value equal to the average temperature across a wide range of thermal fluctuation magnitudes ($0 < m < 0.4$) and frequencies ($1 < \omega < 1000$). The TPC breadth increases monotonically with increasing thermal fluctuation magnitude, while thermal fluctuation frequency only imposes a marginal effect on the TPC breadth (Fig. 4f). Bimodal TPCs emerge under extremely large thermal fluctuation magnitudes ($0.4 < m < 0.5$) (Fig. 4e–f). No J-shaped skew TPCs were observed.

Discussion

Periodic temperature fluctuations drive diverse thermal adaptation patterns, including thermal specialists, generalists, and both unimodal and bimodal thermal performance curves (TPCs). In constant or rapidly fluctuating environments with small magnitudes (Fig. 4a–b), optimal TPC temperatures align with average environmental temperatures, while TPC breadth narrows, resulting in steep, Dirac delta-like TPCs indicating complete specialization. According to theoretical and empirical studies, steeper reaction norms indicate higher performance evolving under slow, predictable fluctuations while greater environmental variability leads to broader TPCs (Amarasekare and Johnson 2017; Schaum et al. 2022). Building on these studies, our results show that higher performance, indicated by narrower TPCs for thermal specialists, can evolve with intermediate thermal magnitudes ($m = 0.25$) and rapid fluctuations ($\omega \geq 1$) when optimal temperatures are below or above the average.

Moderately narrow TPCs, also indicating thermal specialisation, evolve under slow predictable thermal fluctuations ($\omega < 1$), while broader TPCs for thermal generalists evolve under large magnitude ($m > 0.15$) and rapid fluctuations ($\omega > 1$) with optimal temperatures around the average (coloured region; Fig. 4a-d). This highlights the role of environmental fluctuations in promoting the evolution of thermal specialisation vs. generalisation.

Typically, optimal TPC temperatures mismatch average environmental temperatures, and TPC breadth becomes positive, showing non-monotonic patterns across thermal gradients (Fig. 4b, d). Rapid fluctuations increase TPC breadth when optimal temperatures are below the average temperature but decrease when above the average temperature (Fig. 4b). These results agree with previous empirical and theoretical studies on evolutionary adaptation of unimodal TPCs, showing that optimal temperatures can match or mismatch with the average temperature, and the TPC breadth changes under different environmental conditions (Amarasekare and Johnson 2017; Deutsch et al. 2008; Gilchrist 1995; Schaum et al. 2022). These results also align with adaptation patterns of ectotherms from different latitudes (tropical, Mediterranean, and temperate) experiencing seasonal temperature variations. In tropical ectotherms, optimal temperatures match the mean environmental temperatures and TPC breadth is narrow under weak seasonal fluctuations, while in temperate and Mediterranean ectotherms, optimal temperatures exceed the mean environmental temperatures and TPC breadth is wider due to strong seasonal fluctuations (Amarasekare and Johnson 2017; Deutsch et al. 2008; Gilchrist 1995).

Small fluctuations produce narrow TPCs (specialists), while large fluctuations yield moderate TPC breadths (less specialized). When optimal temperatures match average temperatures under large fluctuations, broad TPCs (generalists) evolve. TPC breadth generally increases with fluctuation magnitude and frequency around intermediate average temperatures (Fig. 4b, d). However, rapid intermediate-magnitude fluctuations decrease TPC breadth as frequency increases, especially when optimal temperatures deviate from average (Fig. 4d). Slow fluctuations initially produce wide TPC breadths, narrowing with increasing frequency for specialists that align with average temperature but widening for generalists that mismatch average temperature. Complementing on previous empirical (Foray et al. 2014; Kassen and Bell 1998; Kassen 2002; Schaum et al. 2022) and theoretical (Gabriel and Lynch 1992) studies, our adaptive dynamics model explored the effects of the magnitude of environmental fluctuations in tandem with the frequency of environmental fluctuations to determine the mechanisms of adaptation of the TPCs and resulting performance. If temperature fluctuates slowly (i.e., $\omega < 1$) compared to population dynamics under intermediate magnitudes of environmental fluctuations, this results in a mismatch between the optimal physiological temperature of the TPC and the average environmental temperature (Fig. 4c-d). However, when temperature fluctuations occur rapidly (i.e., $\omega > 1$) with intermediate magnitudes, the population adopts the optimal temperature matching the average (coloured regions; Fig. 4a-b, e-f). In cases of rapid fluctuations of large magnitude, the population adapts to both temperature extremes, resulting in bimodal reaction norms (red regions; Fig. 4a-b, e-f).

Bimodal TPCs emerge under extremely high-magnitude fluctuations around intermediate average temperatures (Fig. 4a-b, e-f), suggesting adaptation to both cold and hot environments, potentially indicating thermal strategy divergence within populations. This does not imply that fitness declines at intermediate temperatures; rather, it indicates that thermal performance is reduced at intermediate temperatures within this thermal regime. Fitness,

instead, is defined as long-term average performance across the entire thermal range, as described in Eq. (2). As an ESS, this bimodal thermal strategy does not represent a minimum in the fitness landscape (Rueffler et al. 2006) but arises from large and rapid temperature oscillations. The flexibility of the beta distribution function allows it to accommodate disruptive selection, leading to the emergence of bimodal TPCs. This suggests that the TPC shape is a population-level strategy, masking disruptive selection below the population level.

Although the model resulted in a single TPC strategy irrespective of whether the environment is constant or fluctuating, this does not contradict many studies on evolutionary adaptation of TPCs that show individual variability even under constant temperature conditions (e.g., Seebacher and Little 2021). This is because Adaptive Dynamics is concerned with the average TPC in the population. In reality, a distribution of TPCs around such average would exist in the population, maintained by drift and sexual recombination. In addition, bimodal TPCs in reality could manifest as a bimodal distribution of unimodal TPCs rather than an actual bimodal TPC. As shown previously (Chevin and Lande 2013), with sufficiently strong frequency-dependent selection, steep thresholds can evolve from continuous sigmoid reaction norms in response to variation in an underlying trait or in a continuous environment, resulting in nearly discrete polyphenism of the expressed trait. This evolutionary response creates a bimodal U-shaped reaction norm, representing a mixture of specialists. Our result indicates how thermal specialists adapt to extreme temperatures at large magnitudes ($m \geq 0.4$) and rapid fluctuations ($\omega > 1$), or to fluctuations of different frequencies when optimal temperatures are around the average (red regions; Fig. 4a–b, e–f). Note that the reaction norm in our model evolved as a continuous bimodal TPC, while it evolved as discrete polyphenisms in Chevin and Lande’s model.

Our adaptive-dynamics model is based on density-dependent competition in different constant and periodic thermal regimes. The model highlights that TPCs in fluctuating environments differ from those observed in the constant environment. This is because the TPC breadth shrinks to zero in constant environments leading to steep TPCs, while it becomes positive leading to narrow TPCs for specialists and possibly even broader TPCs for thermal generalists in fluctuating environments with rapid fluctuations of small or large magnitude. Therefore, the evolution of TPCs in fluctuating environments overall favours thermal generalisation when compared to thermal specialisation in constant environments. Our model emphasizes the importance of considering the average and variability of the thermal environments when examining TPC evolution, since understanding the consequences of climate change on natural populations requires examining how both TPC shapes interact with the thermal regime to determine thermal performance (Estay et al. 2014; Schulte et al. 2011). Local thermal adaptation of complete thermal specialisation of steep TPCs occur due to a constant thermal regime characterised with no to very minimal fluctuations in the ambient temperature. The adaptation of broader unimodal and bimodal TPCs resulted from the periodic thermal regime with sinusoidal fluctuations. An interesting extension to the current model will be to consider multi-species competition at a given temperature, and to investigate the role of environmental variability on the adaptation of TPCs under such interaction. These findings highlight how fluctuation frequency and magnitude complexly shape TPC evolution, demonstrating environmental variability’s influence on ectotherm thermal adaptation strategies.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10682-025-10337-4>.

Acknowledgements A.L.S. acknowledges funding from Federal University Dutse, Jigawa State, Nigeria, under the Tertiary Education Trust Fund (TETFund) 2016 intervention. P.L. and C.H. acknowledge the support from NITeCS (National Institute for Theoretical and computational Sciences) and the National Research Foundation of South Africa (grant 89967). C.H. is also supported by the UK Natural Environment Research Council (NE/V007548/1; GLocal Insect Threat-Response Synthesis, GLiTRS).

Author contributions ALS, conceptualization, analysis and writing; PL and CH, conceptualization, analysis, writing and supervision of the project.

Funding Open access funding provided by Stellenbosch University.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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